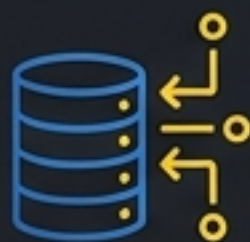




Customer Shopping Behavior Analysis

From Raw Transactions to Strategic Revenue Drivers

Portfolio Project | Python • Azure SQL • Tableau Dashboards

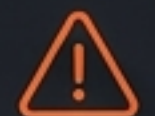


The Dataset

3,900

Purchases across
diverse categories

18 distinct features
per transaction



Quality Flag: **37**
missing values
identified in **Review**
Ratings requiring
statistical imputation



The Dimensions

Demographics:

- Age, Gender, Location,
Subscription Status

Purchases:

- Item, **Category**, **Amount**,
Season, Size, Color

Behavior:

- Discounts, Promo Codes,
Frequency, **Ratings**,
Shipping



The Objective

Uncover spending
patterns, customer
segments, and
subscription
behaviors to guide
**strategic business
decisions.**

Python

Function: Data Extraction
& Feature Engineering

Tech: Pandas, Median
Imputation,
Standardization

Azure SQL Edge

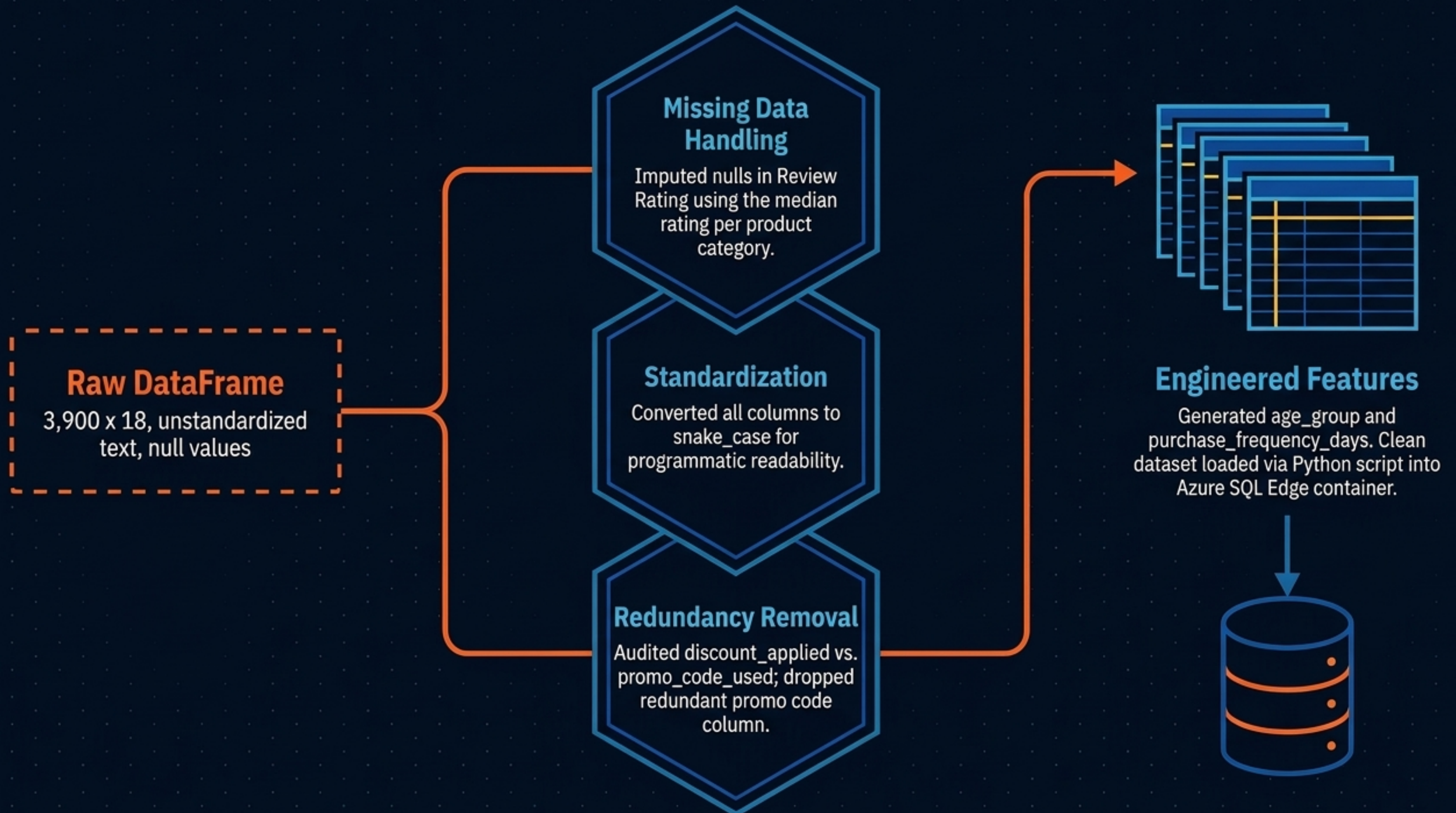
Function: Transaction
Analysis Engine

Tech: Docker Integration,
Relational Structuring,
Multi-dimensional Querying

Tableau / Power BI

Function: Visual Storytelling
& Delivery

Tech: Interactive
Dashboards,
Stakeholder Analytics



Customer Dynamics

Segmentation: Categorizing New, Returning, and Loyal segments based on history.

Subscription Drivers: Contrasting average spend/revenue across subscription statuses and evaluating >5 purchase repeat buyers

Demographics: Calculating total revenue generation by age group and gender.

Product Preferences

Performance Ratings: Identifying the Top 5 products by average review rating.

Category Leaders: Listing the Top 3 most purchased products within each specific category.

Age-Specific Demand: Ranking the Top 5 products and revenue generated per age group.

Revenue & Operations

Margin Risks: Isolating the 5 products with the highest percentage of discounted purchases

Discount Behaviors: Identifying users who apply discounts but still exceed the average purchase amount.

Logistics Impact: Comparing average purchase amounts between Standard vs. Express shipping users.

The Subscriber Premium

Repeat buyers (>5 purchases) demonstrate a high likelihood to convert to paid subscriptions, driving higher baseline revenue.

The Discount Paradox

High-Spending Discount Users rely on discounts to convert but maintain significantly above-average total cart sizes.

Shipping as a Signal

Express shipping selection is highly correlated with stronger overall purchasing behavior, indicating high revenue potential.

MAXIMUM REVENUE GENERATION

Intersection of Subscriber Potential, Discount-Driven Spend, and Strong Purchasing Signals.

TRANSLATING SQL INTO INTERACTIVE ANALYTICS

Bridging the gap between raw backend transaction data and dynamic, interactive Tableau/Power BI dashboards for executive stakeholder storytelling.



OPTIMIZE SUBSCRIPTIONS

Action: Deploy targeted incentives. Offer bag/clothing discounts to attract young adults, and exclusive VIP gifts for high-spending senior jewelry buyers.



TARGETED MARKETING CAMPAIGNS

Action: Reallocate ad spend toward high-revenue age groups and customers favoring express shipping, given their highest purchasing velocity.



MARGIN PROTECTION

Action: Audit discount dependency on the 5 highly-discounted products. Adjust policy to preserve profit margins without stalling sales volume.



PRODUCT POSITIONING

Action: Capitalize on the Top-5 highest-rated items by pushing them to the front lines of marketing campaigns to increase visibility.

